GinCol Lab · Aula

Student worksheet

Physics and Chemistry · 1st year of Bachillerato · The atom and the periodic table · Class 5 of 6 · With Abril

Shells, subshells and the filling order

Class video: https://youtu.be/6bO83Lehono

The code leads to the video.

Name: _____ Date: _____

The video is in Spanish, with English subtitles. On the board, *capa* is shell, *nivel* is level, *subnivel* is subshell and *escalera* is the staircase of energy levels. Z is the atomic number. In Spain the decimal separator is a comma: this sheet uses the decimal point.

Before watching the video

1. In the Bohr model, what are the steps? What happens when an electron drops from one to another?

2. Sodium has eleven electrons. Do you think they all fit in the lowest level? Why?

3. If you take one electron away from a sodium atom, what charge does it have left? How would you write it?

While you watch the video

1. What happens to level 3 on sodium's staircase? Where are levels 1 and 2? Where is the jump that gives the yellow light in your kitchen? Why couldn't that jump exist with just one level for each number?

2. What is a shell? And a subshell? Why is it called a shell? How many subshells does shell 3 have?

3. What is each arrow on the staircase? Where do the letters s, p, d and f come from?

4. What is the difference between an orbit and an orbital? What is each dot in the cloud?

5. Why can't electrons go round like planets?

6. **Fill in:**

Subshell	Orbitals	Electrons
s		
p		
d		
f		

7. Why does sodium's last electron feel a charge of only +1 from far away? Why does 3s drop so much, and 3d hardly at all?

8. Why doesn't sodium's last electron fall to shell 1, as hydrogen's does?

9. What four pieces of data make up an electron's address? Why can't three electrons fit in one orbital?

10. Why is level 3 a single level in hydrogen, while in sodium it splits into three? Use the words "cover" and "slips in".

11. Copy the diagonal rule and use it to write the order of the subshells, up to 5s.

12. Write the configuration of sodium, chlorine and potassium. Why doesn't potassium's electron 19 go to 3d?

13. What does the + sign in Na^+ mean? How many electrons does that ion have?

14. **Your prediction.** If you pull sodium's electrons off one by one, where will the big jumps be? Where were they really? What do those jumps tell you?

15. What can be read from the four peaks of sodium's spectrum?

16. What stayed from the Bohr model, and what was thrown out?

Practice

1. Write the configuration of calcium (Z = 20) and of its ion Ca^{2+}.

2. Write the configuration of the sulfide ion, S^{2-} (sulfur has Z = 16).

3. Which element is $1s^2\ 2s^1$? What does it have in common with sodium and potassium?

Keep thinking

Do orbitals really exist, or are they a drawing that helps us predict? Look for the best argument for each position and say which one you side with, and why.

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